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← Back to Projects

Clarity Coach AI

Production RAG-based educational platform utilizing LLM APIs, vector embeddings, and semantic search for personalized learning experiences

📅 2024-2025 🏆 EITM Grant Recipient </> AI/ML Development

Project Overview

Clarity Coach AI is an AI-powered educational technology application that leverages Retrieval-Augmented Generation (RAG) to deliver personalized, contextually-aware learning experiences. Developed as part of the Entrepreneur in the Making Grant from Scotiabank and NSCC, this project demonstrates advanced capabilities in LLM integration, vector database implementation, and intelligent content retrieval systems.

The platform combines speech analysis technology with automated learning pathway generation, creating adaptive educational content tailored to individual user needs. This dual-purpose application serves both speech pathology clinical use and general educational enhancement.

Problem Statement

Traditional educational platforms lack the ability to dynamically adapt content based on individual learning patterns and speech characteristics. Educators and speech pathologists need tools that can:

- ✓ Provide real-time, contextually relevant feedback

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- ✓ Scale intelligent tutoring without increasing human resource requirements
- ✓ Integrate seamlessly with existing educational workflows

Technical Solution

The application architecture implements a full RAG pipeline with the following components:

- ✓ **Vector Database Integration:** Implemented semantic search using embedding models to retrieve contextually relevant educational content from large knowledge bases
- ✓ **LLM API Integration:** Developed prompt engineering strategies and context management systems for generating personalized responses and learning materials
- ✓ **Automated Learning Pathways:** Built intelligent routing system that adapts content delivery based on user progress, learning style, and performance metrics
- ✓ **Speech Analysis Engine:** Trained and tested machine learning models for speech pattern recognition and analysis
- ✓ **User Interface:** Designed intuitive, accessible interface optimized for educational workflows and real-time interaction

Technologies & Tools

Python	LangChain	OpenAI API	Vector Embeddings		
Pinecone/ChromaDB	scikit-learn	Whisper API	React/Next.js	FastAPI	
PostgreSQL					

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- ✓ Retrieval-Augmented Generation for intelligent content delivery
- ✓ Real-time speech analysis and feedback generation
- ✓ Automated learning pathway creation based on user progress
- ✓ Context-aware prompt engineering for personalized responses
- ✓ Semantic search across educational content libraries
- ✓ Model evaluation and optimization workflows
- ✓ Scalable architecture for production deployment
- ✓ Analytics dashboard for tracking learning outcomes

Development Process

Research Phase: Conducted extensive research on RAG architectures, LLM capabilities, and educational technology best practices. Evaluated various embedding models and vector databases for optimal performance.

Model Training: Trained and tested machine learning models for speech analysis, conducting iterative optimization and validation testing to ensure accuracy and real-world applicability.

RAG Implementation: Built end-to-end RAG pipeline including document processing, embedding generation, vector storage, retrieval mechanisms, and response generation with LLM integration.

User Testing: Conducted user experience testing with educators and students to refine interface design and optimize automated learning pathway logic.

Outcomes & Impact

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- ✓ Secured EITM Grant funding based on technical merit and innovation
- ✓ Created scalable framework applicable to various educational contexts
- ✓ Gained hands-on experience with cutting-edge AI/LLM technologies

Download Full Project Documentation

Access detailed technical documentation, architecture diagrams, and code samples

Download Portfolio (PDF)

Skills Demonstrated

RAG Systems	LLM Integration	Vector Embeddings	Machine Learning
Prompt Engineering	API Development	UI/UX Design	Model Training
System Architecture	Production Deployment		

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